Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 07** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Case Study Question 12(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case Study question of 5 marks with internal choice.
7. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. The height of the tower is: [1]

(A) $7(\sqrt{3}+1)$ m (B) $7(\sqrt{3}-1)$ m (C) $14\sqrt{3}$ m (D) 21 m

Q2. A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°. As the car approaches the tower, the angle of depression: [1]

(A) decreases (B) increases (C) remains constant (D) becomes zero

Q3. If two towers of heights h_1 and h_2 subtend angles of 60° and 30° respectively at the mid-point of the line joining their feet, then the ratio $h_1 : h_2$ is: [1]

(A) 1 : 2 (B) 1 : 3 (C) 3 : 1 (D) $\sqrt{3} : 1$

Q4. If two towers of heights x and y subtend angles of 30° and 60° respectively at the centre of the line joining their feet, then $x : y$ is equal to: [1]

(A) 1 : 3 (B) 3 : 1 (C) $1 : \sqrt{3}$ (D) $\sqrt{3} : 2$

Q5. **Assertion (A):** If both the height of a building and the distance of an observer from its foot are increased by 20%, the angle of elevation of the top of the building remains unchanged.
Reason (R): The trigonometric ratio $\tan\theta = \frac{\text{height}}{\text{distance}}$ is invariant under identical scalar scaling of numerator and denominator. [1]

(A) Both (A) and (R) are true; (R) explains (A). (B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. A tree breaks due to a storm and the broken part bends so that the top of the tree touches the ground making an angle of 30° with it. The distance from the foot of the tree to the point where the top touches the ground is 8 m. Find the total original height of the tree. [2]

Q7. From the top of a cliff 50 m high, the angle of depression of a boat in the sea is 30°. Find the horizontal distance between the boat and the base of the cliff. [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. The length of the shadow of a tower standing on level ground is found to be 21 metres longer when the sun's altitude is 30° than when it was 45°. Prove that the height of the tower is $\frac{21}{\sqrt{3}-1}$ m, and find its numerical value. [Take $\sqrt{3} = 1.732$] [3]

Q9. From a point on a bridge across a river, the bridge is at a height of 15 m above the water level. The angles of depression of the two opposite river banks are observed to be 30° and 45° respectively. Find the perpendicular width of the river. [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

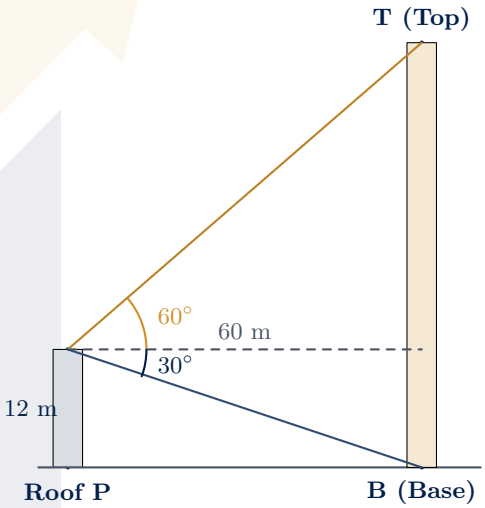
Q10. The angle of elevation of a cloud from a point h metres above the surface of a lake is α and the angle of depression of its reflection in the lake is β. Prove that the height of the cloud above the lake surface is: [5]

$$\frac{h(\tan\beta + \tan\alpha)}{\tan\beta - \tan\alpha}$$

Q11. The angle of elevation of a stationary cloud from a point 60 m above the surface of a lake is 30° and the angle of depression of its reflection in the lake is 60°. Find the height of the cloud above the surface of the lake. [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Observation Deck at Sky Tower Architecture



An architect standing on the flat roof of a commercial building of height 12 m observes a high-rise metropolitan Sky Tower located at a horizontal distance of 60 m.

From the observation vantage point on the building roof (P), the angle of elevation of the top of the Sky Tower (T) is measured to be 60°, while the angle of depression of the base of the Sky Tower (B) is measured to be 30°.

Based on the above mathematical scenario, answer the following questions:

(i) Find the direct line-of-sight distance from the building roof (P) to the base of the Sky Tower (B). [1]

(ii) Determine the height of the section of the Sky Tower (TM) that rises above the horizontal eye-level line of the building roof. [Take $\sqrt{3} = 1.732$] [2]

(iii) Calculate the total vertical height of the Sky Tower (BT). [2]

— OR —

If the architect looks from the base of the building (G_1) towards the top of the Sky Tower (T), write the expression for $\tan\theta$ representing the angle of elevation. [2]

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★